PIXORA AI v2.5.0: COMMAND & CONTROL REFERENCE MANUAL

COMPLETE HOTKEY BINDINGS, TERMINAL EXECUTION SCRIPTS & CONFIGURATION MATRIX

System Version: Pixora AI v2.5.0 | **Target Platforms:** Windows 10/11 (CUDA Enabled), Linux, macOS | **Python Target:** 3.13 | **Hardware:** NVIDIA RTX (CUDA) / CPU Fallback

1. Interactive Live Keyboard Controls (OpenCV Window)

These hotkeys operate live with sub-millisecond dispatch inside the active camera window.

Key	Command	Action & Technical Scope
SPACE	**Typed Sarcasm Capture**	Freezes live video frame; prompts terminal for text; runs multimodal sarcasm fusion (Sfused) & meme generator.
V	**Voice Capture**	Captures face while recording 4s audio at 16kHz PCM via local Whisper STT; routes to sarcasm fusion pipeline.
T	**Toggle Cartoon Preview**	Toggles live real-time cartoon rendering on all detected faces in the video stream.
M	**Cycle Cartoon Mode**	Cycles through 5 modes: avatar_match (Snapchat AR) -> gan -> expression -> cyber_hologram -> opencv.
+ / -	**Avatar Head Scale**	Increases / decreases 3D AR avatar head scale in real time (0.5x to 2.5x, step 0.05x).
P	**Feature Passthrough**	Blends real user eyes and mouth onto cartoon avatar for a hybrid Snapchat filter aesthetic.
R	**Record Looping GIF**	Records a 3-second 30 FPS circular buffer (90 frames) and exports an animated meme GIF to outputs/.
G	**GenAI Scene Meme**	Executes zero-shot Vision-Language Model (VLM) scene context analysis + LLM meme synthesis.
A	**AI Comedy Roast**	Analyzes live biometrics, micro-expressions, and posture to generate a savage AI comedy roast.
S	**Voice Persona (TTS)**	Toggles spoken speech synthesis (SAPI5/pyttsx3) for meme punchlines and AI roasts.
F	**Sci-Fi Sound FX**	Toggles cybernetic interface sound cues (target acquisition chirps, shutter clicks, alert pings).
C	**Cycle Caption Style**	Cycles comedic tone styles: Funny, Sarcastic, Witty, Ironic, Motivational.
D	**Toggle Debug HUD**	Toggles green 478-mesh landmark points, iris gaze reticles, and live FACS cue strength bars.
Q / Esc	**Quit Application**	Safely releases webcam, GPU tensors, audio threads, and terminates cleanly.

2. Application Launch & Web Perception Hub Commands

Target Component	CLI Execution Command	Description & Port
Desktop Cyberpunk AR	`venv\Scripts\python.exe main.py`	Launches primary 30+ FPS OpenCV desktop app with biometrics and Procrustes AR avatar.
Flask Web Perception Hub	`venv\Scripts\python.exe web_app.py`	Starts browser streaming server at http://localhost:5000 with remote controls & graphs.

3. Environment Setup & CUDA Commands

Task	PowerShell / Bash Command
Create venv (Python 3.13)	`py -3.13 -m venv venv && venv\Scripts\Activate.ps1`
Install PyTorch CUDA	`pip install torch torchvision torchaudio --index-url https://download.pytorch.org/whl/cu124`
Verify CUDA GPU	`python -c "import torch; print(torch.cuda.is_available(), torch.cuda.get_device_name(0))"`
Install Core Requirements	`pip install -r requirements.txt`
Install Advanced Requirements	`pip install -r requirements-advanced.txt`

4. Neural Training & Avatar Indexing Commands

All models can be retrained locally on GPU or in Google Colab (Pixora_Online_Colab.ipynb).

Pipeline Component	Execution Script & Parameters	Benchmark & Hardware
30k Avatar Deep Index	`python training/build_cartoon_avatar_index.py`	MobileNetV3 960-D + LAB color indexing across 30k Google Cartoon Set images.
Baseline Sarcasm Model	`python training/train_baseline_local.py`	85.8% Acc / 0.851 F1 on 28.6k headlines (Intel CPU ~1.2s).
Advanced DeBERTa-v3 Sarcasm	`python training/train_advanced_local.py --batch-size 8`	91.4% Acc on held-out test split (RTX 4060 GPU ~4.5 min).
MUStARD Dialogue Adaptation	`python training/finetune_mustard_local.py`	+14.2% in-domain conversational sarcasm accuracy gain.
Custom 4-Block Emotion CNN	`python training/train_emotion_local.py --epochs 30`	68.7% test accuracy on 35.8k FER-2013 facial expression crops.

5. Documentation, Report & Diagram Generators

Artifact / Report	Generator Script	Output File Path
Project Overview PDF	`python generate_overview_pdf.py`	docs/Pixora_AI_Project_Overview.pdf
Command Reference PDF	`python generate_command_reference_pdf.py`	docs/Pixora_AI_Command_Reference.pdf
Draw.io Multi-Tab Architecture	`python generate_drawio_diagram.py`	pixora_architecture.drawio (3 Tabs)

6. Key Configuration Parameters (config.py)

Setting Parameter	Default	Description & Options
CARTOONIFY_MODE	`"avatar_match"`	Active filter: "avatar_match", "gan", "expression", "cyber_hologram", "opencv"
GAN_STYLE	`"portrait_v2"`	AnimeGAN style: "portrait_v2", "portrait_v1", "celeba_distill", "paprika"
CARTOON_AVATAR_SCALE	`1.15`	Head scale factor for Snapchat Procrustes AR avatar alignment.
CARTOON_PASSTHROUGH_EYES	`False`	Blends real user eyes and mouth onto cartoon avatar.
RPPG_HEART_RATE_ENABLED	`True`	Enables contactless optical capillary pulse & stress estimation.
EYE_GAZE_TRACKING_ENABLED	`True`	Enables 3D eye gaze quadrant and laser reticle tracking.
CAPTION_GENERATION_BACKEND	`"local"`	Caption engine: "local" (Qwen2.5-1.5B), "api" (Groq/Gemini), "template"
AUDIO_VOICE_ENABLED	`False`	Spoken TTS voice persona for meme punchlines and AI roasts.

7. Diagnostic & Troubleshooting Commands

Issue / Symptom	Diagnostic & Resolution Command
No video window appearing	`pip uninstall opencv-python-headless -y && pip install opencv-python`
DeepFace / tf-keras error	`pip install tf-keras`
CUDA Out of Memory in Training	`python training/train_advanced_local.py --batch-size 4 --grad-accum 8`
Kill stuck python processes	`Stop-Process -Name python -Force`